Worksheet 12

October 12, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Radians, arc, and sector

A sector has radius 6 cm and central angle $150°$.

(a) Convert the angle to radians.

(b) Find the exact arc length.

(c) Find the exact area of the sector.

Worksheet 12

October 12, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

2. Angles in standard position

Let $\theta = -\frac{5\pi}{4}$.

(a) Convert the angle to degrees.

(b) Sketch the angle in standard position. Label its initial and terminal sides.

(c) Identify the quadrant containing the terminal side.